ELCRO CO.,LTD

– Hook & Loop Fastening Systems for Seat Assembly –

ELCRO ELCRO

Introduction

Company Address

39-1, Hwalcheonsaneop-ro, Duseo-myeon, Ulju-gun, Ulsan

Company Overview

Date of Est.: February 2017

Main Products: Fastener (Hook & Loop)

Key Clients: Hyundai Transys, Hyundai Industrial, FORD, etc.

Phone No.: 052-264-9104~5

CEO: Sang Hee Lee



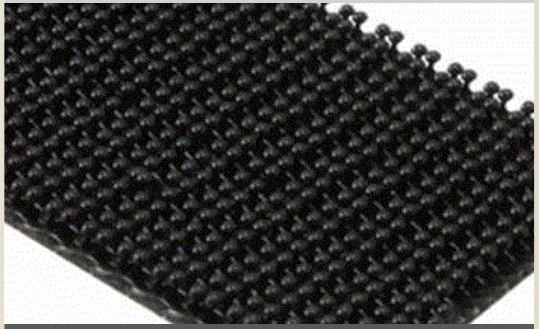


History

Feb. 2017	Developed the widest domestic production method for Mushroom Hook (1440mm)
Apr. 2017	Supply contract with Beijing Das (China)
Aug. 2019	Supply contract with Ford & Changsung Motors
Aug. 2022	Patent application for the Twin Lock manufacturing method
June 2024	Obtained ISO 9001:2015 Quality Management Certification
Oct. 2024	Installation of artificial turf applying Mushroom Hook & development of composite waterproofing method
Jan. 2025	Completed development of Twin Lock (hook-and-hook engaging fastener)
July 2025	Supply contract for Twin Lock in China & India

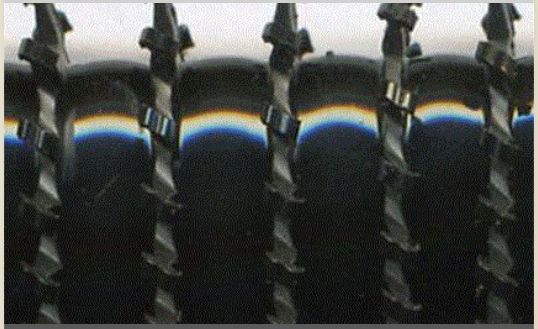
Main Products

Textile Hook



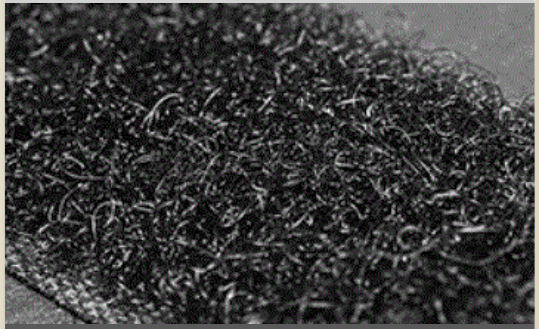
TYPE: J SHAPE

Plastic Hook

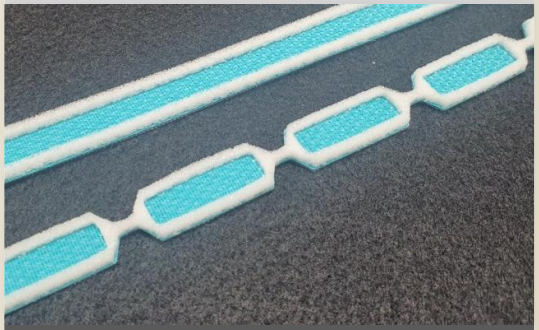


TYPE: T SHAPE

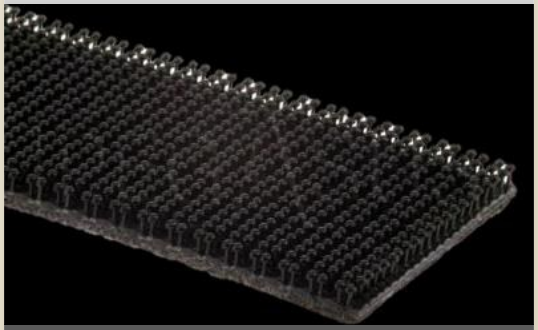
Textile Loop



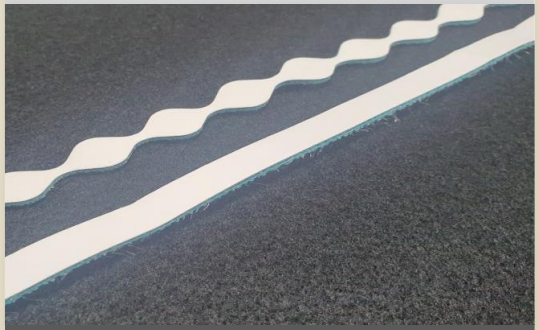
TYPE: TEXTILE LOOP



TYPE: MUSHROOM



TYPE: TWIN-LOCK



TYPE: AUTOMOTIVE LOOP

Main Products

Automotive Interior Materials

HOOK & LOOP



SEAT FOAM / SEAT COVER

TWIN-LOCK



SUNROOF / HEADLINER

Application Status

Application — Seat Assembly



SEAT PAD + HOOK

⟷



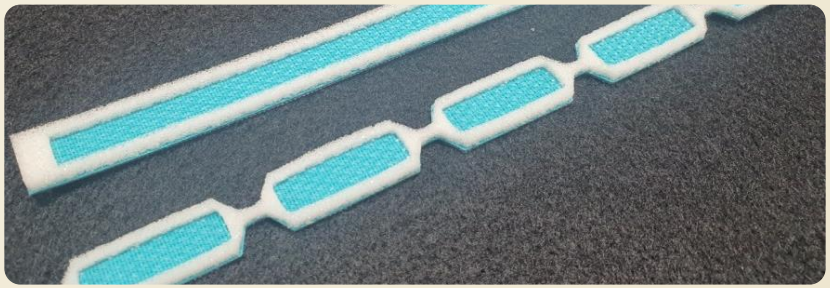
SEAT COVER + LOOP

HOOK (molded into seat pad foam) ⟷ **LOOP** (sewn into seat cover)

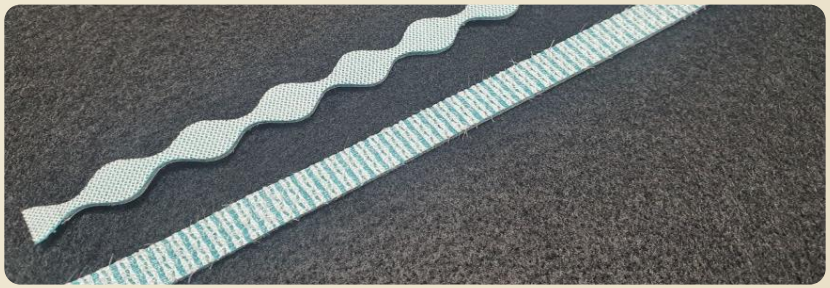
01. HOOK Product Composition (Non-Cover Hook)

Classification	Content	
Composition	HOOK	POLYPROPYLENE (MUSHROOM)
		POLYESTER (BASE)
		PET FILM
		FE POWDER
		POLYESTER (MESH)
Thickness	4.5mm ± 0.1mm	
Tolerance	±0.1mm (LENGTH, WIDTH)	
Separation Strength	HOOK & LOOP	15±3 KGF (standard : 100mm)
	HOOK & SEAT PAD	22±3 KGF (standard : 100mm)

Front



Back Side



02. HOOK Product Composition (Cover Hook)

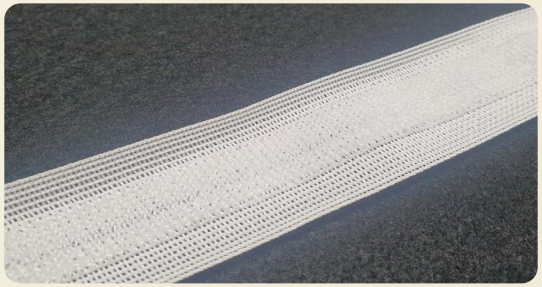
Classification	Content	
Composition	HOOK	POLYPROPYLENE (MUSHROOM)
		POLYESTER (BASE)
		NYLON LOOP
	COVER	PAPER (BASE)
		FE POWDER
		NYLON (LYLEX)
Thickness	4.5mm ± 0.1mm	
Tolerance	±0.1mm (LENGTH, WIDTH)	
Separation Strength	HOOK & LOOP	15±3 KGF (standard : 100mm)
	HOOK & SEAT PAD	22±3 KGF (standard : 100mm)

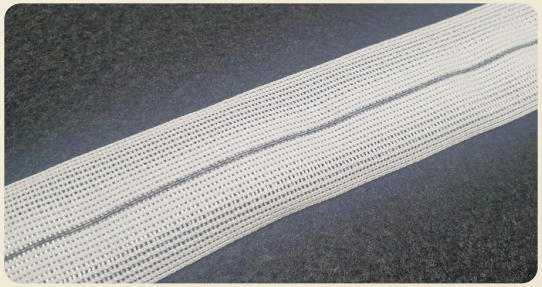
Front

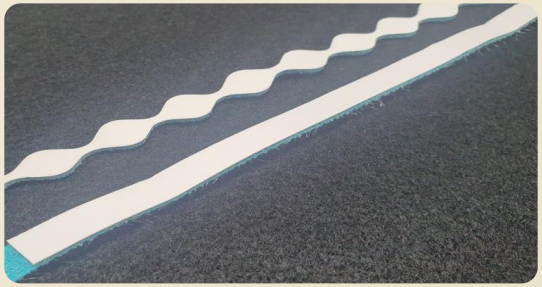
03. LOOP Product Composition

Classification	Content
P/NAME	LOOP
Material	Polyester (Phosphorus-Based Flame Retardant Yarn)
Product Thickness	3.2mm ± 0.1mm
Product Width	40mm / 38mm / 36mm
Separation Strength	15Kgf ± 2Kgf

Front	Side	Back Side







04. HOOK Product Quality Comparison

Advantages	Competitor Products	ELCRO
HOOK & SEAT PAD Integration	 Existing products: approx. 1 mm integrated	 **Approx. 35 mm integration**
Unformed (Void) Prevention Effect	Foaming defects occur	The HOOK rear NYLON LOOP **significantly reduces** foaming defects

#.01 Before / After Foaming #.02 Unformed (Void) Prevention

04-1. HOOK Product Quality Comparison

Advantages	Competitor Products	ELCRO
HOOK and the Rear of HOOK Integration	Separation occurs when removing the cover	**Strong adhesion** — integrated HOOK & rear of HOOK
Occurrence of HOOK Wrinkles	Wrinkles in HOOK cause wrinkles during seat cover assembly	**No wrinkles** in HOOK

#.03 Rear Integration #.04 Wrinkle Prevention

05. HOOK Separation Strength

Unit : Kgf | Width 10mm × Length 100mm | Vertical separation as measured by P/P Gauge

No.	Test	Standard	Test Method	Result	Note
1	**HOOK & LOOP**	Width 10mm × Length 100mm	Vertical separation as measured by P/P Gauge	**17**	
2	**HOOK & PAD**			**23**	

06. HOOK Reliability Test

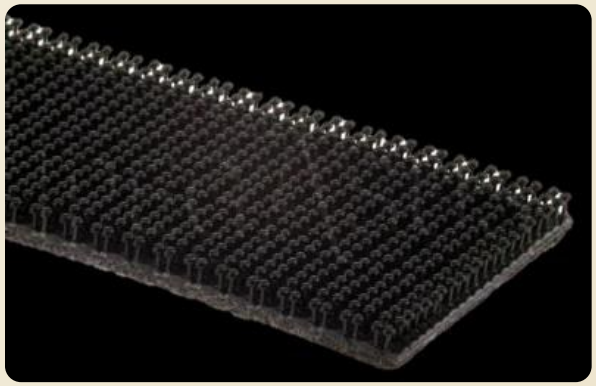
NO	Test Title	Test Standard	Testing Requirements	Test Results
1	**Inflammability of Interiors**	MS 300-08	Burn Velocity 80mm/min below	**Approved**
2	**Smell of Interiors**	MS 300-34	3 grade over	**Approved**
3	**Volatile Organic Compounds**	MS 300-55	TVOC – 2ppm less HCHO – 2ppm less	**Approved**
4	**Regulate Heavy Metal**	MS 201-02	Pb, Hg, Cd, Cr6+ No Detected	**Approved**

MS: HYUNDAI-KIA Motors Engineering Standard
(Consistent with the motor safety standards in North America and Europe)

Twin Lock

01. TWIN LOCK Product Specification Sheet

Classification	Content
Product Name	**TWIN LOCK**
Composition Layer	A POLYOLEFIN
	B ACRYLIC FOAM
	C ACRYLIC RESIN (Adhesive)
	D Tape Cover (Release Paper)
Color	BLACK
Thickness	3.6mm ± 0.2mm
Combined Thickness	5.6mm ± 0.3mm
No. of Protrusions	400 (Horizontal: 20, Vertical: 20)



Standard: 25mm × 25mm
Front / Back

Reliability Test: Flame Retardancy 54mm/min — Pass | Combustibility Grade 1.5 — Pass | Hazardous Substances Not Detected — Pass | Tensile Strength 20–28 KGF

02. Twin Lock Separation Strength Test

Unit : Kgf | Width 25mm × Length 25mm | Instron 5582

1. #400 & #400

	#.1	#.2	#.3	#.4
23°C	25.1	26.8	27.5	25.8
-30°C	33.2	32.4	35.5	33.7
120°C	8.3	9.5	8.3	8.7

2. #400 & #250

	#.1	#.2	#.3	#.4
23°C	20.8	18.4	19.5	20.2
-30°C	28.7	25.6	23.4	28.2
120°C	7.4	7.6	7.1	7.7

03. Twin Lock & Primer Heat Resistance Test

Width 25mm × Length 25mm | HMC Test specification

1. #400 & #400

Test Method	Result
1. Chamber Temperature: 100°C (Oven) 2. Weight: 1 kg SPCC Steel Plate & 2 kg SPCC Steel Plate 3. Specimen: RAIL (AL) 4. Apply PRIMER (9960 BTX) and attach — test conducted 1 day later 5. 8-Hour Test	**APPROVED**

2. #400 & #250

Test Method	Result
1. Chamber Temperature: 100°C (Oven) 2. Weight: 1 kg SPCC Steel Plate & 2 kg SPCC Steel Plate 3. Specimen: RAIL (AL) 4. Apply PRIMER (9960 BTX) and attach — test conducted 1 day later 5. 8-Hour Test	**APPROVED**

04. TWIN LOCK & PRIMER 3-CYCLE Test

Width 25mm × Length 25mm | HMC Test specification | 1. #400 & #250

Test Method	PRIMER
3-Cycle / 1-Cycle standard: High temperature (120°C) for 8 hours → Low temperature (-40°C) for 8 hours → Room temperature (23°C / 50% RH) for 8 hours	520K — **APPROVED** 4298UV — **APPROVED** 9960BTX — **APPROVED**

Product	SIDE MOLDING	SIDE MOLDING	RAIL
PRIMER	520K	4298UV	9960BTX
Result	**PASS**	**PASS**	**PASS**

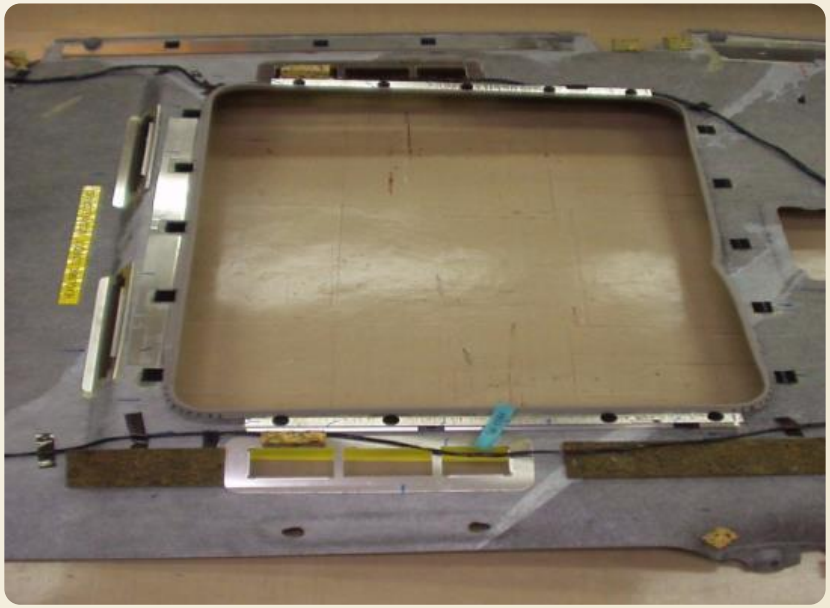
05. TWIN LOCK Quality Certification

MS: Hyundai & Kia Motors Technical Standards
(Compliant with North American and European Automotive Safety Standards)

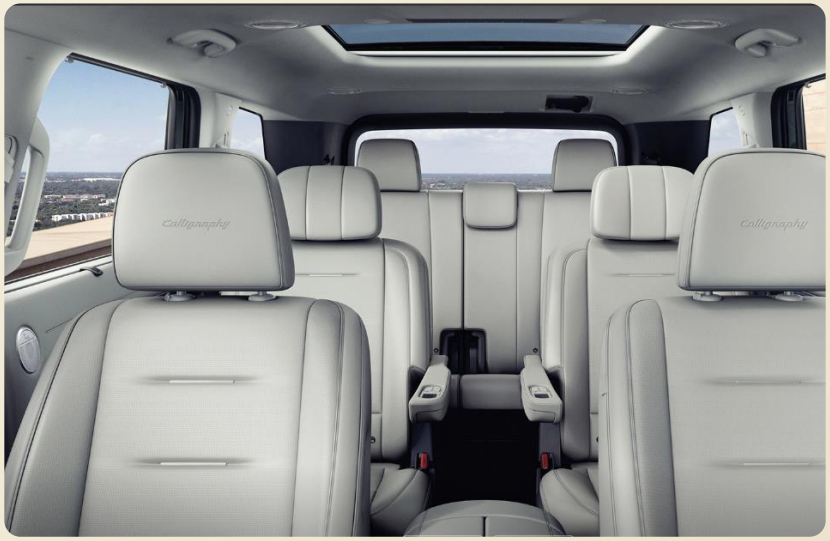
NO	Test Title	Test Standards	Test Requirements	Remarks
1	**Combustibility of Interior Materials**	MS 300-08	Combustion Speed 80 mm/min or Less	**Pass**
2	**Odor of Interior Materials**	MS 300-34	Grade 3 or Higher	**Pass**
3	**Harmful Substances Prohibition & Reporting**	MS 201-02		**Pass**

06. TWIN LOCK Application Status

• SUNROOF & HEADLINER • QTR GLASS • DOOR TRIM



SUNROOF FRAME



SEAT / INTERIOR

Equipment Status

- Thermoforming Machine
- Slitting Machine
- Lamination / Die Cutting Machine
- Extrusion Coating Machine
- MUSHROOM Forming Equipment
- Fabric Specifications Cutting Equipment
- Finished Product Specifications Cutting
- Film / Fabric Extrusion Coating Equipment
- Non-Cover Machine #1
- Non-Cover Machine #2
- Double Forming Machine
- Six-Fold Forming Machine
- Non-Cover Production Equipment
- Twin-Lock Molding Equipment
- Twin-Lock Post-Processing Equipment

Thank You.